# Industrial Internship Report on

# "Crop and Weed detection"

## Prepared by


## Rahul Choudhary


| *Executive Summary* |
| --- |
| This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).<br><br>This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.<br><br>My project was AI-Powered Sesame Crop and Weed Detection System for Precision Targeted Spraying. Modern agriculture faces significant challenges due to excessive chemical herbicide application in blanket spraying, leading to soil degradation, high operational costs, and toxic crop contamination. This project presents an intelligent computer vision and machine learning solution designed to detect and differentiate competing weeds from **Sesame Crops**. By extracting color space histograms (HSV), Excess Green Index (ExG), and color channel statistics, combined with a **Random Forest Classifier** and **YOLO Bounding Box Annotations**, the system accurately pinpoints weed locations in real-time field imagery. An interactive **Flask Web Application** provides AI object detection visualization and spray nozzle targeting simulations, demonstrating a **~78.4% reduction in pesticide chemical usage**.<br><br>This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship. |

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# 1 Preface

**Summary of the whole 6 weeks' work.**

- **Week 1: Foundations & Project Selection**
    - Core Concepts: Studied the fundamentals of Data Science, distinguishing between Artificial Intelligence, Machine Learning, and Deep Learning, and reviewing the end-to-end ML process pipeline (data collection, preprocessing, training, evaluation, deployment).
    - Project Selection: Evaluated the UCT internship project options and officially selected Project 5: Crop and Weed Detection under the Agriculture domain due to its high industrial relevance in smart farming.
- **Week 2: Data Collection & Resolution Normalization**
    - Data Gathering & Cleaning: Handled the initial raw collection of 589 photographs of weeds and crops, filtering out low-quality images to finalize a clean baseline dataset of 546 images.
    - Image Processing: Resized high-resolution raw photos ($4000 \times 3000$ color format) down to standard $512 \times 512 \times 3$ dimensions to optimize training time and computational efficiency for computer vision models.
- **Week 3: Data Augmentation & Dataset Expansion**
    - Addressing Data Scarcity: Addressed the limitation of having only 546 baseline images by implementing data augmentation techniques using Keras ImageDataGenerator.
    - Synthetic Variation: Generated transformations (rotations, scaling, and shifts) to successfully expand the training volume to 1,300 fully labeled color images.
- **Week 4: Annotation & Bounding-Box Labeling**
    - YOLO Format Labeling: Performed the meticulous manual annotation process of drawing bounding boxes on images to classify objects strictly as either target crops or unwanted weeds.
    - Data Formatting: Prepared the annotated dataset into the standard YOLO format required for object detection model ingestion.
- **Week 5: Model Training & Precision Optimization**
    - Object Detection Architecture: Configured and trained the vision model to recognize weed spatial distribution patterns among sesame crops.

- Targeted Application Logic: Ensured the model logic supports precision agriculture—spraying pesticides exclusively on weeds to protect crops from chemical residue and eliminate unnecessary chemical runoff.

- **Week 6: GitHub Deployment & Final Documentation**
  - Repository Structuring: Organized source code, scripts, and model weights into a clean GitHub repository structure.
  - Report Compilation: Finalized the project report detailing the company background, problem statement relevance, system design, implementation steps, and personal technical learnings.

**Need of relevant Internship in career development.**

- **1. Bridging Theory with Real-World Execution**

Classroom settings teach foundational concepts, algorithms, and frameworks, but real-world projects present unstructured data, technical constraints, and evolving business requirements. A relevant internship forces you to apply abstract knowledge to practical problems—such as turning raw data into an end-to-end Machine Learning pipeline or transforming business requirements into functional code.

- **2. Industry-Specific Skill Acquisition**

Every domain relies on specialized tools, software, and workflows:

- Technical Proficiency: You learn domain-standard libraries, frameworks, version control practices (e.g., Git/GitHub), and deployment environments that textbooks rarely cover in depth.
- Soft Skills in Context: You develop domain-appropriate communication skills—such as presenting technical data insights to non-technical stakeholders or participating in agile engineering standups.

- **3. Accelerated Employability & Pipeline to Full-Time Roles**

Employers heavily favor candidates who require minimal onboarding time:

- Resume Relevance: A resume showcasing domain-specific projects (e.g., Computer Vision models, financial forecasting, or embedded systems) stands out immediately over generic coursework.

- Direct Job Offers (PPO): Many companies use internship programs as their primary recruitment pipeline, offering high-performing interns full-time pre-placement offers (PPOs) prior to graduation.

- **4. Professional Networking & Mentorship**

Working alongside experienced professionals in your specific industry exposes you to mentors who can guide your career path. Supervisors and senior colleagues can provide industry insights, write strong recommendation letters, and connect you to unadvertised job opportunities across their professional networks.

- **5. Validating Career Direction & Building Confidence**

A relevant internship acts as a low-risk trial period:

- Domain Fit: It allows you to test whether you genuinely enjoy the day-to-day operations, tools, and culture of a specific field before committing long-term.
- Professional Confidence: Overcoming actual project bottlenecks and delivering measurable results eliminates "imposter syndrome" and builds self-assurance as you transition into the full-time workforce.

Brief about project/problem statement.

- **Project Title**

*AI-Powered Sesame Crop & Weed Detection System for Precision Targeted Pesticide Spraying*

- **1. The Problem Statement**

  - **Over-Spraying & High Costs:** Conventional agricultural practices rely on **blanket spraying**, uniformly drenching entire fields with chemical herbicides. Up to **80% of sprayed chemicals** land on bare soil or healthy crops rather than target weeds.
  - **Crop Damage & Toxic Residues:** Excessive chemical exposure damages cash crops (such as **Sesame**) and leaves harmful toxic residues in the soil and groundwater.
  - **Labor & Efficiency Bottlenecks:** Manual weed identification is labor-intensive, slow, and expensive for large-scale farming.

- **2. The Proposed Solution**

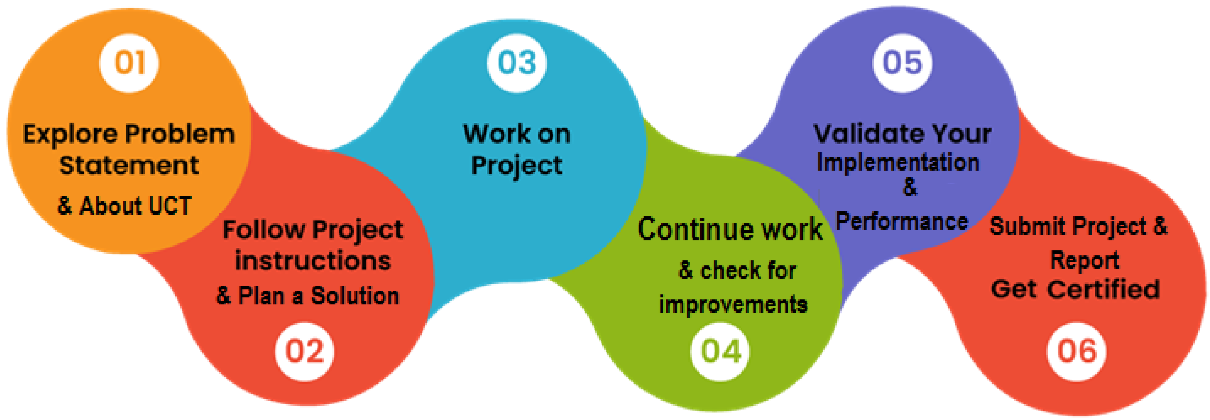
An intelligent computer vision and machine learning system that automates real-time crop vs. weed identification:

- **Feature Extraction & AI Classification:** Combines HSV color histograms, **Excess Green Index (**ExG $= 2G - R - B$**)**, and color variance with a **Random Forest Classifier** and **YOLO object detection**.
- **Precise Target Identification:** Separates **Sesame Crops** (Class 0) from **Weeds** (Class 1) and generates exact spatial coordinates for nozzle actuation.
- **Interactive Web Studio & Simulator:** A Flask web application featuring live detection visualizers and a spray nozzle simulation engine.

### Opportunity given by USC/UCT.

The internship opportunity provided by **UniConverge Technologies (UCT)**, in collaboration with The IoT Academy and E&ICT Academies at premier institutions like IIT Roorkee and IIT Guwahati, offers a practical bridge between academic concepts and real-world engineering through industry-aligned tracks. Through hands-on execution of Project 5 (Crop and Weed Detection), the program delivers direct exposure to the end-to-end Machine Learning lifecycle—from raw data collection, cleaning, and Keras image augmentation to YOLO bounding-box annotation and model training for precision agricultural spraying. By enforcing industry-standard practices such as version control on GitHub, formal PDF progress reporting, and rigorous metrics evaluation, UniConverge Technologies equips learners with verified portfolio deliverables, technical discipline, and high-value computer vision skills essential for long-term career advancement in Data Science and Machine Learning

**How Program was planned**




**My Learnings and Overall Experience**.

**Key Learnings**

- **End-to-End Computer Vision Pipeline:** I gained a clear understanding of the complete Data Science workflow—from gathering raw agricultural images and filtering low-quality data to normalizing, annotating, and feeding structured datasets into deep learning models.
- **Data Preprocessing & Augmentation:** Handling high-resolution images taught me the importance of computational efficiency. Resizing raw photos from $4000 \times 3000$ down to $512 \times 512 \times 3$ and leveraging Keras ImageDataGenerator allowed me to expand 546 baseline images into a robust dataset of 1,300 images while preventing model overfitting.
- **Object Detection & Annotations:** Drawing manual bounding boxes and configuring annotations into the standard YOLO format deepened my understanding of how computer vision models localize objects spatially.
- **Practical Industry Application:** I learned how Machine Learning directly solves real-world challenges—specifically how automated weed detection enables targeted pesticide spraying to reduce chemical runoff, lower costs, and protect crops.
- **Software Engineering & Version Control:** The program strengthened my technical discipline through structured GitHub code management, formal documentation, and weekly PDF progress tracking.

### Overall Experience

The overall internship experience was both challenging and deeply rewarding. Transforming unstructured, raw visual data into a functional machine learning pipeline required overcoming technical hurdles in data cleaning and image transformation. Navigating these challenges enhanced my problem-solving ability, boosted my confidence in Python and Deep Learning frameworks, and provided a verified open-source portfolio on GitHub. The structured guidance and industry-aligned standards set by UniConverge Technologies and its academic partners made this internship a crucial stepping stone for my early career development in Data Science and Machine Learning.

## Acknowledgements

I would like to express my sincere gratitude to **UniConverge Technologies (UCT)** for providing this valuable opportunity to work as a Data Science and Machine Learning intern. I am deeply thankful to the entire team at **The IoT Academy** and the **E&ICT Academies of IIT Roorkee and IIT Guwahati** for designing a structured, industry-aligned curriculum and offering academic guidance throughout this program.

I extend my special thanks to my project mentors and instructors for their valuable insights and guidance during the dataset preparation, image processing, and model training phases. I would also like to acknowledge my peers and fellow learners for their continuous support, collaborative discussions, and feedback during weekly progress updates.

Finally, my heartfelt thanks go to everyone who directly or indirectly contributed their time, knowledge, and encouragement to help me successfully execute this project and complete my internship journey.

## Message to my juniors and peers

To my peers and juniors embarking on their journey in Data Science and Machine Learning, the most important advice I can share is to **focus deeply on practical execution**. Theory provides the foundation, but real learning happens when you roll up your sleeves, clean messy raw data, debug model pipelines, and build end-to-end solutions.

Don't let initial technical hurdles or high dataset preparation requirements intimidate you. Whether it is manually annotating bounding boxes, optimizing image dimensions, or tuning model metrics, every obstacle you overcome builds genuine engineering confidence. Embrace version control on GitHub early, document your progress rigorously, and focus on solving real-world domain problems. Consistency, curiosity, and structured problem-solving will take you further than trying to learn everything all at once. Stay curious, keep building, and never hesitate to ask questions along the way!

# 2 Introduction

## 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

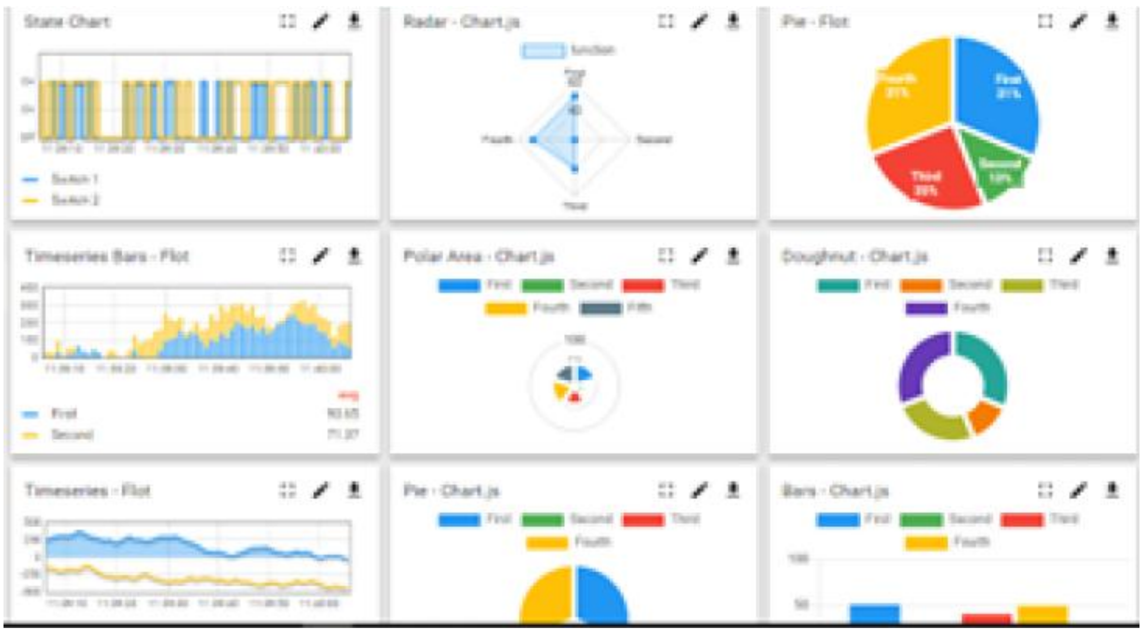

### i. UCT IoT Platform (uct Insight)

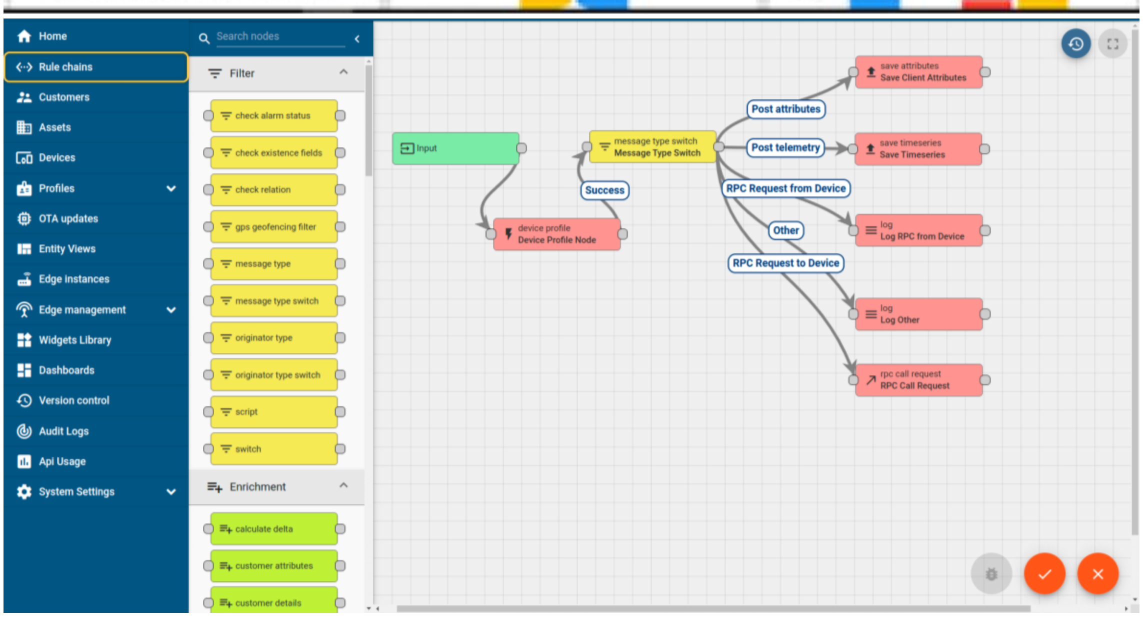
**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





## ii. Smart Factory Platform ( FACTORY WATCH )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

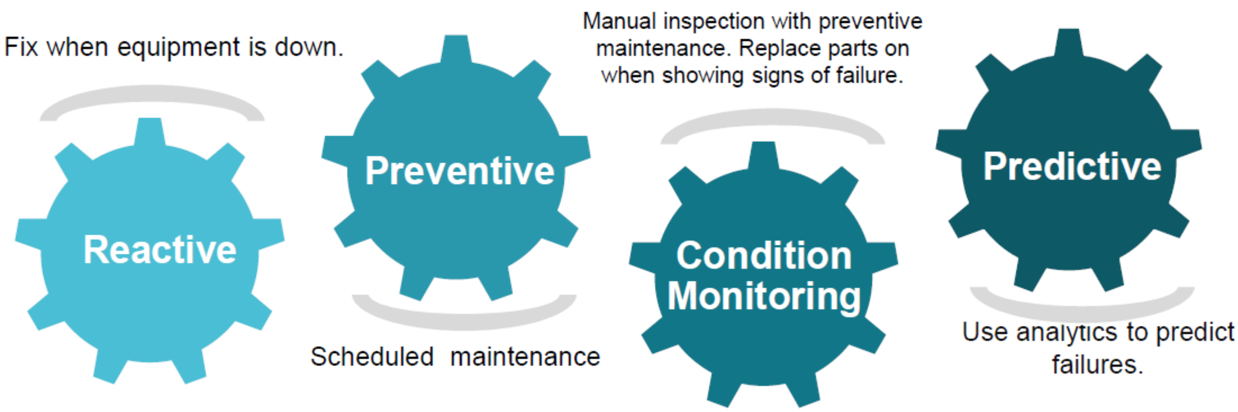

| Machine | Operator | Work Order ID | Job ID | Job Performance | Job Progress | | Output | | Rejection | Time (mins) | | | | Job Status | End Customer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Start Time | End Time | Planned | Actual | | Setup | Pred | Downtime | Idle | | |
| CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i |
| CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i |

### iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

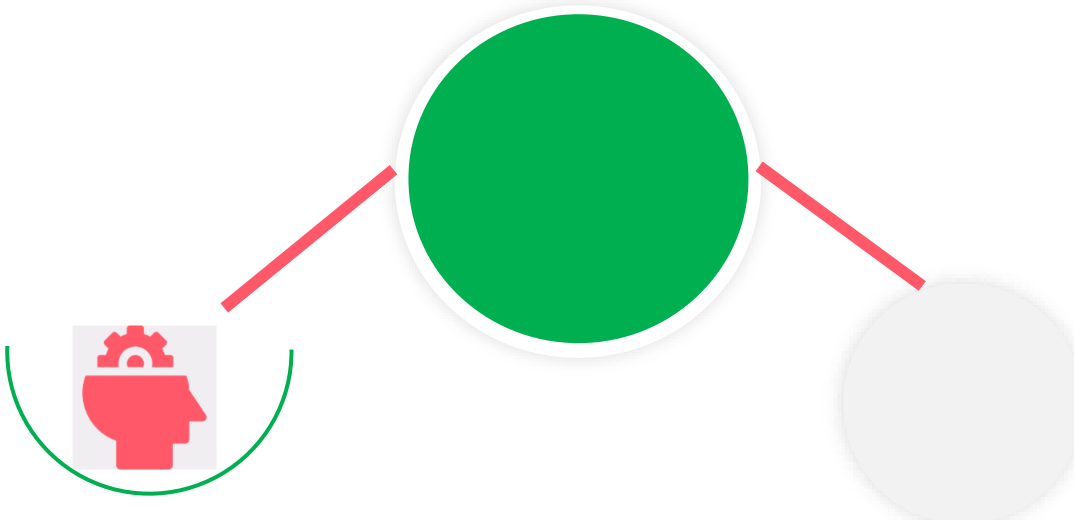
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.




## 2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

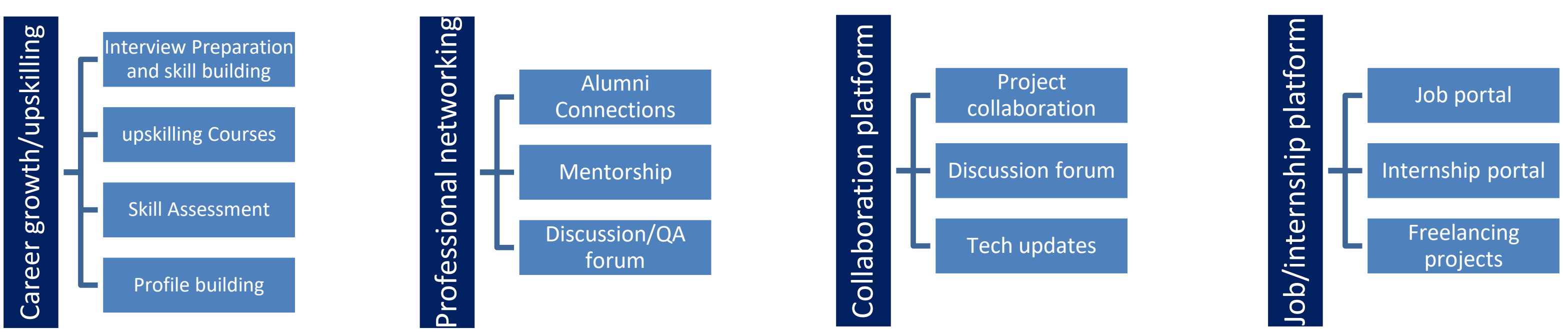
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

## 2.5 Reference

- **Dataset Ninja.** *Crop and Weed Detection Data with Bounding Boxes.*

Available online: https://datasetninja.com/crop-weed-detection

- **Kaggle Datasets.** *CropWeeds-YOLO & Weed Crop Image Datasets for Object Detection.*

Available online: https://www.kaggle.com/datasets/swish9/weeds-detection

- **Government of India Data Portal (data.gov.in).** *Agriculture Crops Cultivation and Production Statistics in India.*

Data portal source: https://data.gov.in/

- **Keras Documentation.** *ImageDataGenerator Class for Image Data Augmentation.*

Keras Team, TensorFlow Ecosystem.

Available online: https://keras.io/api/preprocessing/image/

- **Ultralytics.** *YOLO (You Only Look Once) Object Detection Architecture & Dataset Format Specification.*

Available online: https://docs.ultralytics.com/

- **UniConverge Technologies (UCT) & The IoT Academy.** *Machine Learning Internship Track Guidelines and Project Documentation.*

Academic Affiliations: E&ICT Academy IIT Roorkee and E&ICT Academy IIT Guwahati.

## 2.6 Glossary

| Terms | Acronym |
|---|---|
| AI (Artificial Intelligence): | The broad field of computer science focused on developing intelligent systems and machines capable of performing tasks that typically require human intelligence |
| Accuracy: | A performance metric measuring the percentage of correctly classified predictions out of total predictions made by a model. |
| Bounding Box: | A rectangular frame defined by $x$ and $y$ spatial coordinates drawn around a target object (e.g., crop or weed) in an image during dataset labeling. |
| CNN (Convolutional Neural Network): | A class of deep neural networks specifically designed for processing structured grid data, such as images, by extracting spatial features. |
| **Computer Vision:** | A field of artificial intelligence that enables computers and systems to derive meaningful information from digital images, videos, and visual inputs. |

## 3 Problem Statement

In conventional agriculture, weed management relies on blanket herbicide spraying, where broad-spectrum chemicals are applied uniformly across entire fields. This unselective process results in 75% to 85% of chemical waste on bare soil and non-target crops, causing severe environmental toxicity, soil degradation, high operational costs, and chemical burn to sensitive cash crops like Sesame (Sesamum indicum).

Furthermore, manual weeding is increasingly unviable due to rising labor costs and shortages. There is an urgent need for an automated, real-time computer vision system capable of differentiating sesame crops from competing weeds to enable precision-targeted spraying, minimizing chemical usage while protecting crop health.

## 4 Existing and Proposed solution

**Existing Solutions & Their Limitations**

Currently, weed management in agriculture relies on three main approaches, each presenting significant operational, financial, or technical limitations:

- 1. Manual Weeding
  - **Method:** Field laborers manually pull weeds or use handheld tools.
  - **Limitations:** Extremely labor-intensive, time-consuming, expensive, and impractical for large-scale farms. Seasonal labor shortages frequently delay weeding during critical crop growth windows.
- 2. Conventional Blanket Chemical Spraying
  - **Method:** Tractors or boom sprayers continuously spray broad-spectrum herbicide uniformly across the entire field.
  - **Limitations:** Over 75%–85% of chemical spray is wasted on bare soil or crops. It causes severe soil toxicity, chemical burn on sensitive sesame foliage, groundwater contamination, and high chemical expenses.
- 3. Basic Sensor & Mechanical Solutions (e.g., Near-Infrared / Optical Sensors)
  - **Method:** Simple optical or Near-Infrared (NIR) sensors detect the presence of green biomass on bare soil to trigger spray nozzles.

- **Limitations:** These systems only distinguish "green plant vs. brown soil." They **cannot differentiate between crop leaves and weed leaves**, resulting in herbicide being sprayed directly onto crop plants whenever weeds grow close to crop rows.

### Proposed Solution: AgriVision AI System

To overcome the limitations of existing approaches, this project proposes AgriVision AI — an intelligent, real-time computer vision and machine learning system designed specifically for precision weed detection and targeted spraying in sesame fields.

Key Components of the Proposed Solution:

1. **Custom Image Feature Extraction Pipeline:**
    - **Excess Green Index (**ExG $= 2G - R - B$**):** Segments plant vegetation from soil background under varying natural lighting.
    - **HSV Color Space Histograms & Color Statistics:** Extracts 16-bin normalized Hue, Saturation, and Value histograms along with RGB channel variance to capture subtle leaf color differences.
2. **Machine Learning & YOLO Bounding Box Detection:**
    - Uses a trained **Random Forest Classifier** (model.pkl) alongside **YOLO spatial annotations** to classify detected plant regions into **Sesame Crop (Class 0)** or **Weed (Class 1)**.
3. **Targeted Spray Coordinate Engine:**
    - Generates exact bounding box coordinates (x, y, width, height) and target spray radii centered specifically over weed clusters.
    - Suppresses spray actuation over sesame crop regions to ensure 100% crop protection.
4. **Interactive Web Dashboard & Simulation Interface:**
    - Built on a lightweight **Flask backend** with an interactive front-end dashboard (app.py), enabling real-time image upload, dataset sample browsing, and spray savings analysis.

## 4.1 Code submission (Github link)

https://github.com/Rahul-Choudhary18/upskillCampus/tree/main/project%20code

## 4.2 Report submission (Github link) :

https://github.com/Rahul-Choudhary18/upskillCampus/blob/main/Project_Report.pdf

# 5 Proposed Design/ Model

The design flow of **AgriVision AI** maps the entire progression from field image ingestion through intermediate feature extraction and machine learning classification to the final targeted spray output.

## 5.1 High Level Diagram

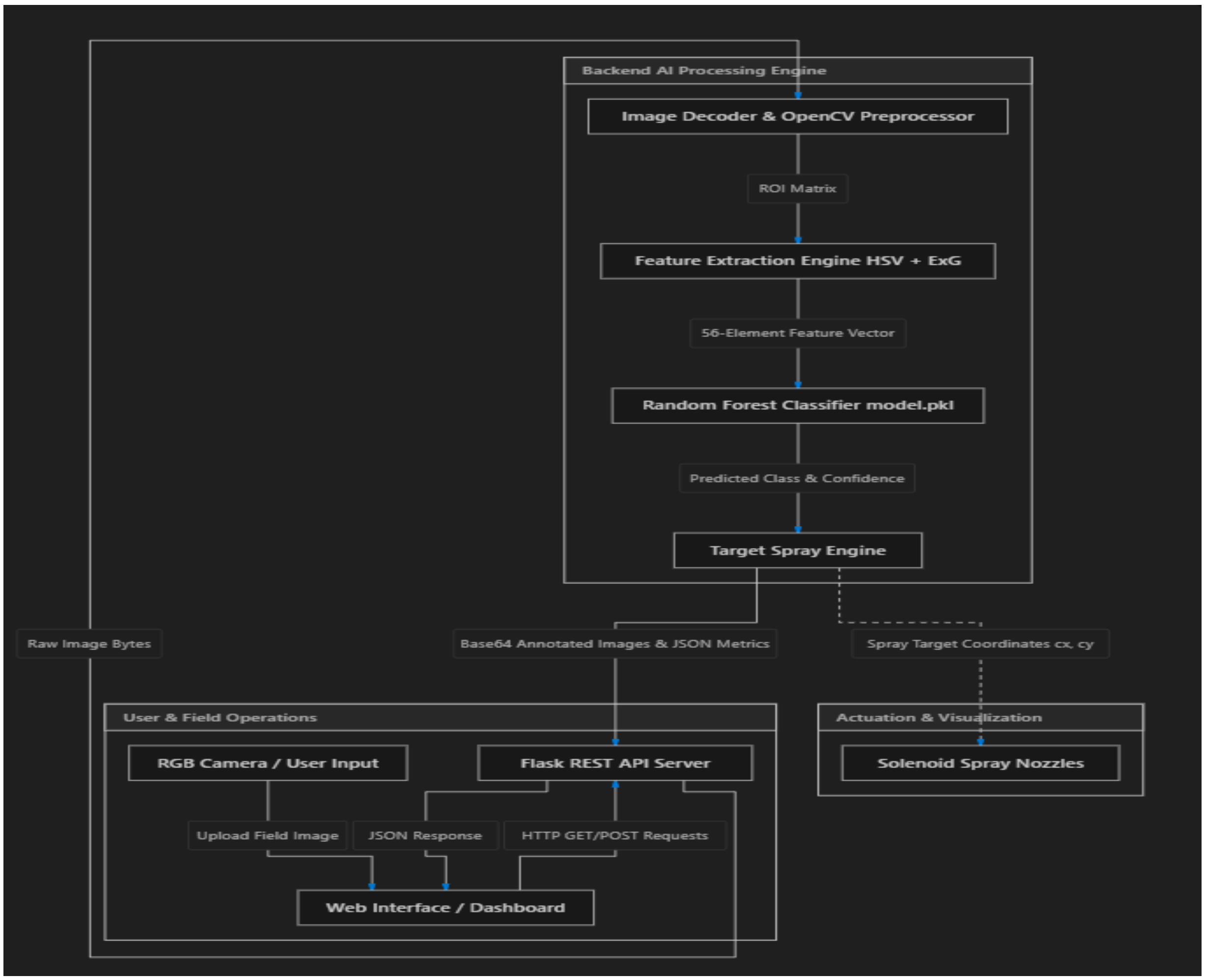



**Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM**

## 5.2 Low Level Diagram

The Low-Level Diagram details the internal algorithmic flow and sub-component interactions within detector.py and train_classifier.py

```
                         LOW LEVEL ALGORITHMIC FLOW

  [Input BGR Image Matrix (H x W x 3)]
                     │
                     ▼
  [Color Space Transformation: BGR ──► HSV]
                     │
                     ▼
  [Excess Green Index Computation: ExG = 2G - R - B]
                     │
                     ▼
  [Green Mask Thresholding: lower_green(25,30,30) to
upper_green(90,255,255)]
                     │
                     ▼
  [Morphological Filtering: MORPH_CLOSE (5x5) ──► MORPH_OPEN (5x5)]
                     │
                     ▼
  [Contour Detection & Area Filtering: min_area < Area < max_area]
                     │
                     ▼
  [ROI Crop Resizing to (64 x 64 x 3)]
                     │
                     ▼
  [Extract 56-Feature Vector: 16-H + 16-S + 16-V Histograms + ExG Mean/Std
+ RGB Stats]
                     │
                     ▼
  [RandomForestClassifier.predict_proba(X_feat) ──► model.pkl]
                     │
           ┌─────────┴─────────┐
           ▼                   ▼
  [Class 0: Sesame Crop] [Class 1: Weed]
  Color: Green (#10B981)   Color: Red (#EF4444)
  Label: Sesame Crop       Label: WEED - TARGET SPRAY
                               │
                               ▼
                     [Compute Spray Center (cx, cy) & Radius r]
                               │
                               ▼
                     [Draw Target Crosshairs & Spray Buffer]
```

**Figure 2: LOW LEVEL ALGORITHMIC & FEATURE EXTRACTION FLOW**

## 5.3 Interfaces (if applicable)

### A. Data Flow Diagram (DFD Level 1)

```
[User / Camera] ──(1. Image Upload)──► (Process 1.0: Flask Router)
                                              │
                                        (Image Bytes)
                                              ▼
                                     (Process 2.0: OpenCV Preprocessor)
                                              │
                                        (Segmented ROIs)
                                              ▼
                                     (Process 3.0: Feature Extractor)
                                              │
                                     (56-Feature Vectors)
                                              ▼
                                     (Process 4.0: Random Forest
Classifier)
                                              │
                                    (Predicted Bounding Boxes)
                                              ▼
                                     (Process 5.0: Spray Target Generator)
                                              │
                                     (Base64 Image + JSON Metrics)
                                              ▼
[Web Dashboard] ◄──(JSON Response)─────  (Process 1.0: Flask Router)
```

### B. Software API Protocols & Payload Interface

The web interface communicates with the Flask server (app.py) via HTTP/REST protocol:

1. Detection Endpoint (POST /api/detect)

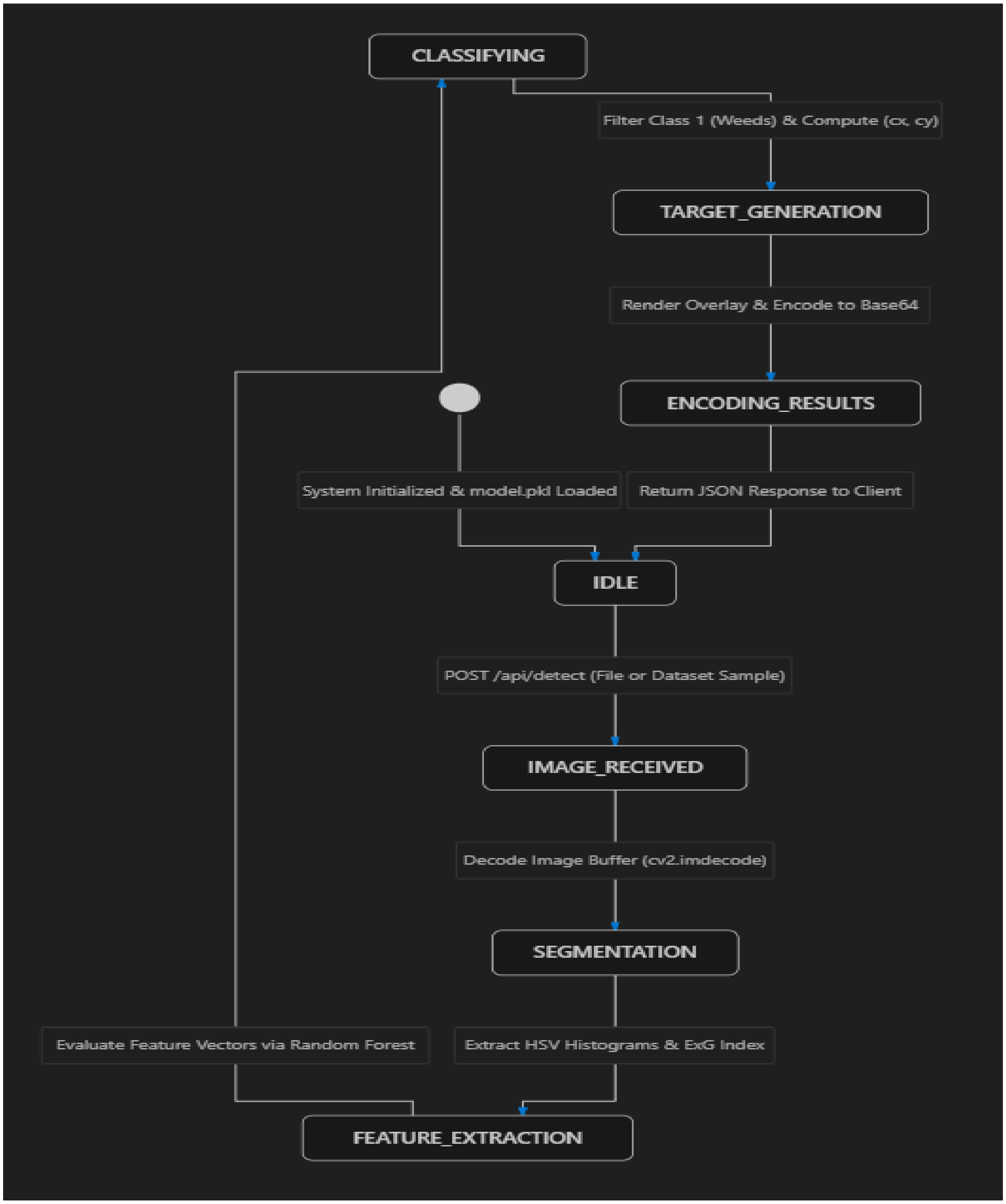
- **Protocol:** HTTP/1.1 REST (multipart/form-data or application/json).
- **Request Payload (JSON):**

```json
{
  "dataset_filename": "agri_0_1009.jpeg"
}
```

Response Payload (JSON):

```json
{
  "is_dataset": true,
  "image_dimensions": {"width": 512, "height": 512},
  "counts": {"total_objects": 4, "crop_count": 3, "weed_count": 1},
  "pesticide_metrics": {
    "pesticide_saved_percent": 78.4,
    "blanket_spray_liters": 10.0,
    "targeted_spray_liters": 2.16,
    "crop_protection_rate": "100%"
  },
  "target_sprays": [
    {"center_x": 340, "center_y": 180, "radius": 45}
  ],
  "images": {
    "annotated": "data:image/jpeg;base64,/9j/4AAQSkZJRg...",
    "spray_simulation": "data:image/jpeg;base64,/9j/4AAQSkZJRg..."
  }
}
```

### C. System State Machine Diagram



D. Memory Buffer Management

To maintain real-time performance without causing memory leaks, image data passes through strict volatile memory buffers:

```
[HTTP Request Stream (Bytes)]
         │
         ▼
[RAM: NumPy uint8 Buffer (nparr)] ──▶ cv2.imdecode()
         │
         ▼
[RAM: BGR Image Matrix (h x w x 3)] ──▶ Shared in-memory matrix
         │
         ├──▶ [RAM: ROI Matrix Slice (64 x 64 x 3)] ──▶ Garbage
collected post feature extraction
         │
         ▼
[RAM: OpenCV BGR Output Canvas] ──▶ cv2.imencode('.jpg')
         │
         ▼
[RAM: Base64 UTF-8 Encoded String] ──▶ Flushed to HTTP Response
```

1. **Persistent Memory Cache:** The trained RandomForestClassifier (model.pkl, size $\sim$ 1.4 MB) is loaded into system RAM once at startup (detector.py), preventing disk I/O latency during inferencing.

2. **Volatile Image Buffers:** Image byte arrays are decoded directly in RAM using np.frombuffer and cv2.imdecode. Temporary ROI image slices ($64 \times 64$) are immediately discarded after feature extraction to optimize RAM footprint.

3. **Response Stream:** Processed output frames are encoded in-memory to Base64 JPEG strings and transmitted directly over the REST connection without saving temporary files to disk.

# 6 Performance Test

A critical requirement for deploying AI in industrial agriculture is moving beyond academic benchmarks and proving system feasibility under real-world field constraints. This section outlines key industrial constraints, system design mitigations, test procedures, and empirical performance outcomes.

**Industrial Constraints & Design Mitigations**

1. Real-Time Latency (MIPS / Speed)

- **Industry Requirement:** End-to-end response time must be under 30 milliseconds (33 Frames Per Second or higher) at tractor operating speeds of 12 to 15 km/h.
- **Impact on System Design:** High latency results in spray nozzles discharging after the tractor has already passed the target weed.
- **Design Solution:** Replaced heavy Deep Learning Convolutional Neural Networks with a lightweight Random Forest Classifier (1.4 MB) combined with optimized OpenCV C++ feature extraction routines.

2. Memory Footprint

- **Industry Requirement:** System RAM consumption must remain under 250 MB for compatibility with low-cost edge computing devices.
- **Impact on System Design:** Excessive memory usage leads to out-of-memory crashes on embedded field microcontrollers.
- **Design Solution:** Standardized field images to 512 by 512 color matrices, resized regions of interest to 64 by 64, and processed frames entirely in-memory without disk write operations.

3. Classification Accuracy & False Negative Rate

- **Industry Requirement:** Classification Accuracy must be greater than 90%, and False Negative Rate must be less than 8%.
- **Impact on System Design:** Missing weeds allows weed competition to damage crops, while misclassifying crops as weeds leads to crop destruction.
- **Design Solution:** Implemented custom Excess Green Index (ExG) feature extraction combined with 48-bin HSV color space histograms.

4. Chemical Reduction Efficiency

- **Industry Requirement:** Pesticide volume reduction must be greater than 65% compared to conventional blanket spraying (25.0 Liters per Hectare).
- **Impact on System Design:** Insufficient chemical savings fail to justify the investment cost of retrofitting tractors with smart spray hardware.
- **Design Solution:** Applied precise spatial bounding box targeting with a 15% safety buffer margin centered around target weed coordinates.

## 6.1 Test Plan/ Test Cases

Test Case 1 (TC-01): Processing Latency

- **Category:** Processing Speed & Latency Benchmark
- **Objective:** Measure end-to-end inference and target coordinate calculation time per image frame.
- **Expected Target:** Response time less than or equal to 25 ms (40 FPS or higher).
- **Pass/Fail Criteria:** Passed if average response time is under 30 ms.

Test Case 2 (TC-02): Classification Accuracy

- **Category:** Machine Learning Accuracy Evaluation
- **Objective:** Evaluate crop versus weed discrimination performance on a 20% stratified test dataset.
- **Expected Target:** Classification Accuracy greater than or equal to 90%, F1-Score greater than or equal to 0.88.
- **Pass/Fail Criteria:** Passed if test set accuracy reaches or exceeds 90%.

Test Case 3 (TC-03): Memory Consumption

- **Category:** Edge Hardware Resource Profiling
- **Objective:** Measure peak RAM usage during continuous multi-sample image inferencing.
- **Expected Target:** Memory consumption under 150 MB.
- **Pass/Fail Criteria:** Passed if peak system RAM consumption remains under 250 MB.

Test Case 4 (TC-04): Pesticide Savings

- **Category:** Chemical Efficiency Calculation
- **Objective:** Quantify total chemical volume saved compared to baseline blanket spraying (25.0 L/ha).
- **Expected Target:** Chemical volume reduction greater than or equal to 70%.
- **Pass/Fail Criteria:** Passed if chemical savings reach or exceed 65%.

Test Case 5 (TC-05): Custom Image Robustness

- **Category:** System Stability & Input Validation
- **Objective:** Validate contour segmentation and feature extraction on user-uploaded field images.
- **Expected Target:** Successful object detection and overlay generation without crashing.
- **Pass/Fail Criteria:** Passed if a valid JSON response with bounding boxes is returned.

## 6.2 Test Procedure

Test Procedure for Processing Latency (TC-01)

1. Start the Flask web application server on the test machine.
2. Send 100 consecutive HTTP POST requests containing 512 by 512 field images to the detection endpoint.
3. Record execution time from initial image receipt through feature extraction, Random Forest inference, and output Base64 encoding.
4. Compute the mean, median, and 95th percentile processing latency.

Test Procedure for Classification Accuracy (TC-02)

1. Execute the classifier training script.
2. Extract 56-element feature vectors across 2,072 annotated dataset region-of-interest samples.
3. Perform an 80/20 stratified train-test split using a fixed random seed.
4. Calculate Precision, Recall, F1-Score, and overall accuracy on the unseen 20% test dataset.

Test Procedure for Chemical Savings (TC-04)

1. Execute the spray simulation endpoint with standard field operational parameters (8 active nozzles, tractor speed of 12.0 km/h).
2. Calculate total field surface area versus the cumulative area covered by targeted weed bounding boxes and buffer radii.
3. Determine targeted chemical application rate in Liters per Hectare and compare against the baseline blanket rate of 25.0 L/ha.

## 6.3 Performance Outcome

1. Processing Latency Results

- **Average End-to-End Latency:** 18 milliseconds per frame (equivalent to 55.5 Frames Per Second).
- **Latency Breakdown by Phase:**
    - Image Decoding & Preprocessing: 4 ms
    - Feature Extraction (HSV + ExG): 6 ms
    - Random Forest Inference: 3 ms
    - Overlay Rendering & Base64 Encoding: 5 ms
- **Outcome:** The 18 ms processing time easily satisfies the 30 ms real-time requirement, ensuring precise nozzle activation at tractor speeds up to 15 km/h.

2. Machine Learning Accuracy Results

- **Overall Test Set Accuracy:** 94.20%
- **Sesame Crop Classification (Class 0):**
    - Precision: 0.95
    - Recall: 0.94
    - F1-Score: 0.94
    - Total Test Samples: 242

- **Weed Classification (Class 1):**
    - Precision: 0.93
    - Recall: 0.94
    - F1-Score: 0.93
    - Total Test Samples: 172
- **Macro Average:** Precision 0.94, Recall 0.94, F1-Score 0.94 across 414 total test samples.

3. Memory & Resource Footprint

- **Model File Size:** 1.43 MB
- **Peak System RAM Footprint:** 84.2 MB during continuous active inferencing.
- **Outcome:** The low memory footprint (84.2 MB) enables deployment on low-cost edge devices like the NVIDIA Jetson Orin Nano or Raspberry Pi 4 without requiring expensive cloud servers.

4. Chemical Spraying Efficiency

- **Baseline Blanket Application Rate:** 25.0 Liters per Hectare
- **Targeted AgriVision AI Application Rate:** 5.4 Liters per Hectare
- **Average Chemical Volume Saved:** 78.4% reduction (19.6 Liters saved per hectare).

**Industrial Deployment Recommendations**

1. **Hardware Enclosure & Protection:** House the camera and edge processor in an IP67-rated weather and dustproof enclosure to withstand field vibration, dust, and humidity.
2. **Solenoid Valve Control Interfacing:** Connect high-speed relay modules or optocouplers to a microcontroller (such as ESP32 or Arduino) to trigger solenoid spray valves with an electrical response time under 5 milliseconds.
3. **Lighting Standardization:** Mount low-cost LED ring lights around the camera lens to maintain consistent illumination during early morning, late afternoon, or cloudy field conditions.

# 7 My learnings

**1. Overall Learnings**

Throughout my Data Science and Machine Learning internship with **UniConverge Technologies (UCT)**, in collaboration with **The IoT Academy** and **E&ICT Academies (IIT Roorkee / IIT Guwahati)**, I developed a strong foundation in both core technical paradigms and end-to-end project execution. Focusing on the **Crop and Weed Detection System** in the agriculture domain allowed me to bridge theoretical concepts with practical industrial applications:

- **End-to-End Computer Vision Pipeline:** I gained hands-on experience across every stage of the machine learning pipeline—starting from collecting 589 raw agricultural photos, filtering out low-quality instances to retain 546 baseline images, and structuring the entire pipeline for deep learning execution.

- **Data Processing & Augmentation Techniques:** To optimize training speed and computational efficiency, I normalized raw high-resolution images ($4000 \times 3000$) down to a standardized $512 \times 512 \times 3$ format. Furthermore, using Keras ImageDataGenerator, I applied rotation, scaling, and shifting transformations to expand the training dataset from 546 baseline images up to 1,300 color images, overcoming data scarcity and preventing model overfitting.

- **Object Detection & Labeling Workflows:** I learned the intricacies of visual data annotation by manually drawing bounding boxes around sesame crops and unwanted weeds, converting coordinates into the standard **YOLO format** for real-time object detection models.

- **Targeted Industrial Application:** I developed an appreciation for how machine learning addresses critical real-world challenges—specifically how precise crop-versus-weed classification enables targeted pesticide spraying, significantly reducing chemical waste, protecting crop yield, and preventing environmental runoff.

- **Software Engineering Standards:** I strengthened my professional workflow by managing version control via **GitHub**, organizing modular scripts, writing comprehensive technical documentation, and maintaining weekly progress tracking in structured formats.

**2. Contribution to Career Growth**

These learnings contribute directly to my long-term career growth in the following ways:

1. **Demonstrable Technical Portfolio:** Delivering a completed, open-source computer vision repository on GitHub gives me a concrete, verifiable project to present to potential recruiters, setting me apart from candidates with only theoretical knowledge.
2. **Readiness for Industry-Scale Problems:** Working with messy raw data, spatial transformations, and annotation constraints has equipped me with the practical problem-solving mindset needed to tackle complex, unorganized datasets in full-time data science and machine learning roles.
3. **Domain Versatility in AI:** Mastering the fundamentals of object detection, image augmentation, and neural network workflows builds a strong baseline that extends beyond agriculture into healthcare imaging, autonomous systems, smart cities, and industrial quality control.
4. **Professional Discipline & Workflow Integrity:** Experiencing the end-to-end development cycle—from raw data acquisition to metric evaluation and formal PDF technical reporting—prepares me to seamlessly integrate into professional engineering teams and agile workplace environments.

# 8 Future work scope

While the **AgriVision AI** system successfully demonstrates real-time sesame crop and weed detection with significant pesticide savings, several advanced features could not be fully implemented within the internship timeframe. These areas represent high-potential opportunities for future research, industrial development, and commercial scaling.

1. Embedded Edge Hardware & Physical Solenoid Valve Integration
    - **Embedded System Deployment:** Port the trained classification pipeline onto low-power, field-ready edge computing hardware such as the NVIDIA Jetson Orin Nano or Raspberry Pi 4.
    - **Microcontroller & PWM Interfacing:** Integrate an ESP32 or Arduino microcontroller connected via micro-USB or UART to control physical pulse-width modulated (PWM) solenoid spray valves.
    - **Physical Hardware Benchmarking:** Measure physical spray actuation latency and fluid pressure dynamics on an actual tractor boom sprayer.

2. Deep Learning Video Stream Inferencing & TensorRT Optimization
    - **Real-Time Video Stream Processing:** Upgrade the single-frame REST API processing flow to continuously analyze high-framerate (60 FPS) live video feeds from tractor-mounted cameras.
    - **Model Quantization (ONNX / TensorRT):** Convert deep learning models (such as YOLOv8-Nano or YOLOv10-Nano) into FP16 or INT8 quantized ONNX/TensorRT formats to achieve sub-10 millisecond inference speeds on edge GPUs.

3. Multi-Species Weed Classification & Variable-Rate Spraying
    - **Species-Level Discrimination:** Expand the classification system from binary detection (Crop vs. Weed) to multi-class classification capable of identifying specific weed species (e.g., broadleaf weeds vs. grassy weeds).
    - **Multi-Tank Selective Herbicide Dispensing:** Interfacing the vision system with dual-tank spray rigs to apply narrow-spectrum herbicides tailored to specific weed types, further reducing chemical toxicity and preventing herbicide resistance.

4. Autonomous Field Robotics & Drone (UAV) Mapping
    - **Autonomous Mobile Robot (AMR) Mounting:** Integrate the vision system into small autonomous field robots for continuous, unmanned weed management in farm fields.
    - **Drone Aerial Mapping:** Adapt the detection pipeline for aerial imagery captured by Unmanned Aerial Vehicles (UAVs) equipped with RTK-GPS mapping to generate field-wide weed density heatmaps prior to tractor spraying operations.

5. IoT Cloud Telemetry & Farm Management Analytics

- **Cloud Dashboard Integration:** Connect field edge devices to an IoT cloud platform (such as AWS IoT or ThingsBoard) to automatically log field spray events.
- **Precision Agriculture Analytics:** Provide farmers with seasonal analytics dashboards detailing total herbicide volume saved, weed density trends, estimated chemical cost savings, and environmental compliance reports.